



Measures of Variability

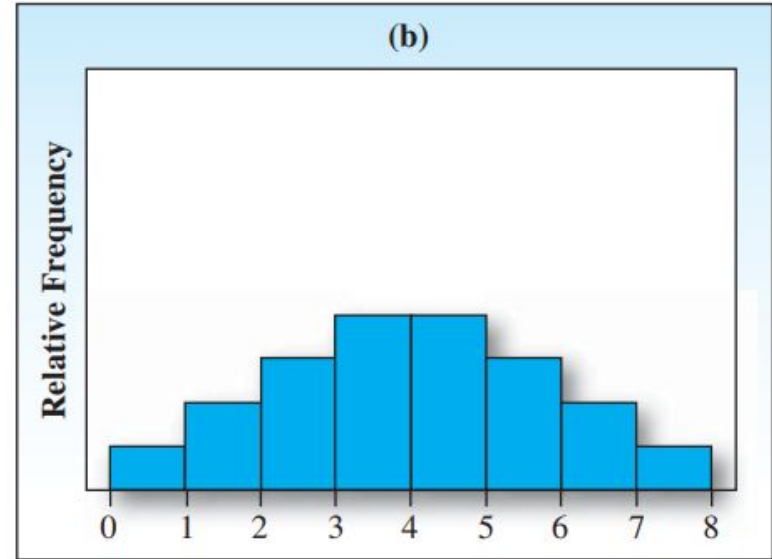
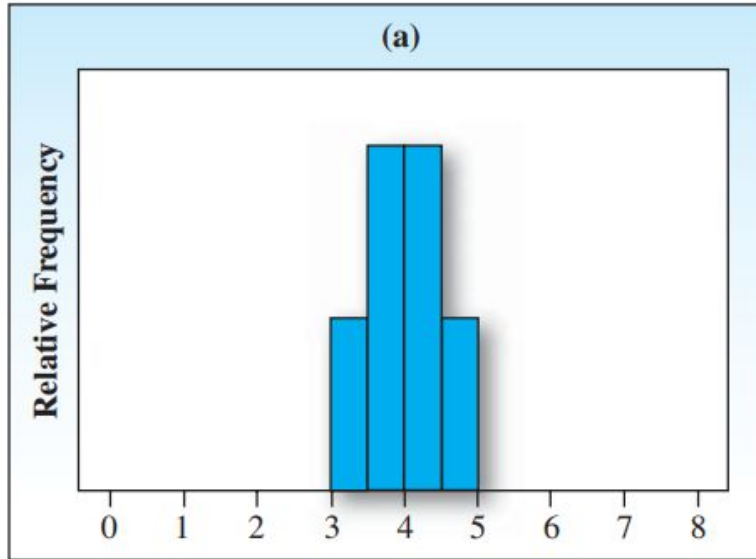


What is measures of variability

Some questions

- What is variable?
- What are the types of variables?
- Why do we tend to find the center of data? Give some examples

What is measures of variability





What is measures of variability

Data sets may have the same center but look different because of the way the numbers spread out from the center.



Variability or Dispersion

Variability, or dispersion, refers to how spread out or clustered the values in a dataset are around a **central value**, such as the mean or median.

Understanding variability helps us assess the consistency of data and provides insight into how typical or unusual certain data points may be.

Measures of variability can help you create a mental picture of the spread of the data. The simplest measure of variation is the ***range***.



Range

The range, R , of a set of n measurements is defined as the difference between the largest and smallest measurements.



Practice

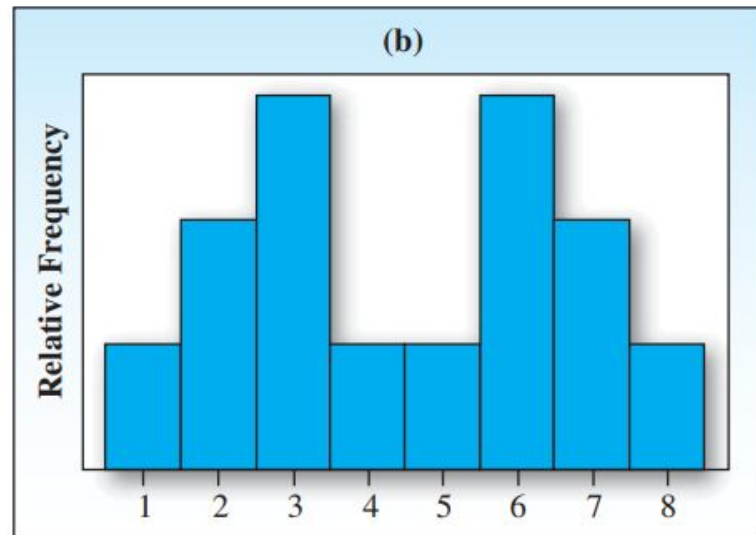
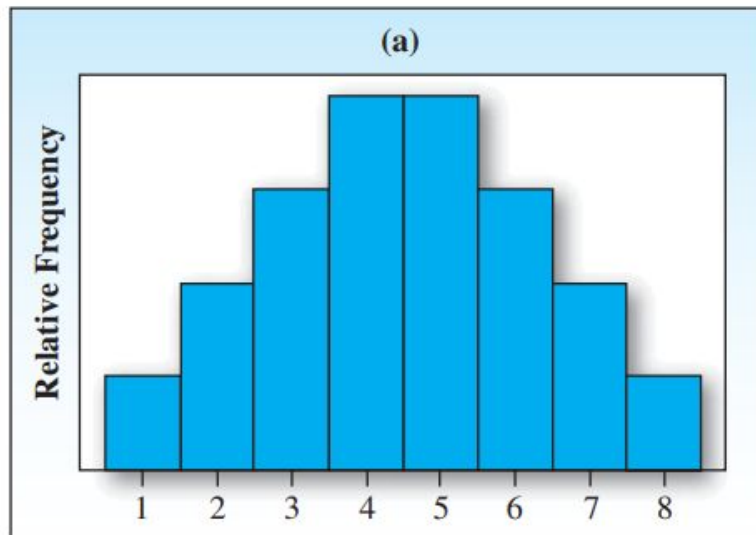
Birth Weights of 30 Full-Term Newborn Babies

7.2	7.8	6.8	6.2	8.2
8.0	8.2	5.6	8.6	7.1
8.2	7.7	7.5	7.2	7.7
5.8	6.8	6.8	8.5	7.5
6.1	7.9	9.4	9.0	7.8
8.5	9.0	7.7	6.7	7.7



Birth Weight Data

- What is the *Range*?
- How many classes should we create?
- Can we draw a frequency histogram?
- Method of left inclusion



**Is there a measure of variability that is
more sensitive than the range?**



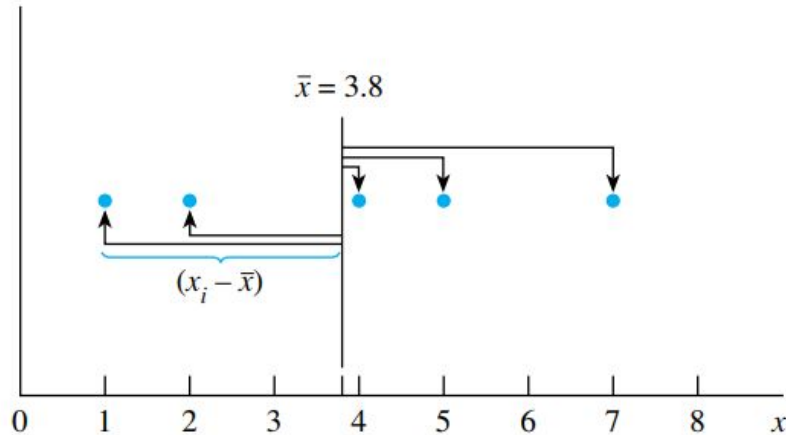
Measure of Variability

Let's draw a diagram with the following data

5, 7, 1, 2, 4

Mean: $19/5 = 3.8$

Measure of Variability



The horizontal distances between each dot (measurement) and the mean $\bar{x}$ will help you to measure the variability.

If the distances are large, the data are more spread out or variable than if the distances are small.



Deviation

If x_i is a particular dot (measurement), then the ***deviation*** of that measurement from the mean is $(x_i - \bar{x})$.

Measurements to the right of the mean produce positive deviations, and those to the left produce negative deviations.

Deviation

Computation of $\Sigma(x_i - \bar{x})^2$

x	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
5	1.2	1.44
7	3.2	10.24
1	-2.8	7.84
2	-1.8	3.24
4	.2	.04
19	0.0	22.80

Because the deviations in the second column of the table contain information on variability, one way to combine the five deviations into one numerical measure is to average them. The average will not work because some of the deviations are positive, some are negative, and the sum is always zero.



Deviation

Another possibility might be to disregard the signs of the deviations and calculate the average of their absolute values. This method has been used as a measure of variability in exploratory data analysis and in the analysis of **time series data**.



Variance

From the sum of squared deviations, a single measure called the variance is calculated (Column 3).

Variance is the mean of the squared deviations from the mean, showing how data points are spread out.



Variance

The variance of a **population** of N measurements is the average of the squares of the deviations of the measurements about their mean μ . The population

variance is denoted by σ^2 and is given by the formula

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$



Variance

The variance of a sample of n measurements is the sum of the squared deviations of the measurements about their mean $\bar{x}$ divided by $(n - 1)$. The sample variance is denoted by s^2 and is given by the formula

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$



Standard Deviation

The standard deviation of a set of measurements is equal to the positive square root of the variance.



Tchebysheff's Theorem

Given a number k greater than or equal to 1 and a set of n measurements, at least $[1 - (1/k^2)]$ of the measurements will lie within k standard deviations of their mean.